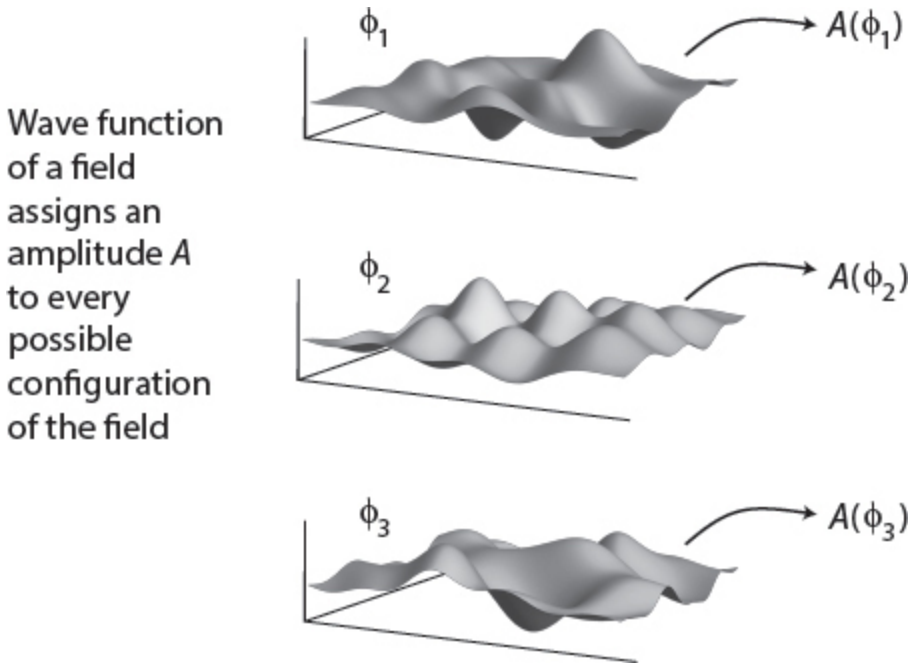


This is the difference between a classical field and a quantum wave function. A classical field is a function of space, and a classical theory with many fields would describe multiple functions of space overlapping with one another. The wave function in quantum field theory is not a function of space, it's a function of the set of all configurations of all the classical fields. (In the Core Theory, that would include the gravitational field, the electromagnetic field, the fields for the various subatomic particles, and so on.) An intimidating beast, but something physicists have learned to understand and even cherish.

All of this implicitly assumes the Many-Worlds version of quantum mechanics. We didn't say anything about decoherence and branching, but we have been taking for granted that all we really need is a quantum wave function and an appropriate version of the Schrödinger equation, and the rest will take care of itself. That's exactly the Everettian situation. (Sometimes when people say "the Schrödinger equation" they are referring specifically to the version Schrödinger originally wrote down, which is only appropriate for non-relativistic point particles, but there's no difficulty in finding a version of the equation for relativistic quantum fields or any other system with a Hamiltonian.) In other theories, one often needs additional variables or rules about how wave functions spontaneously collapse. When we move to field theory, it's not immediately clear what those extra ingredients should be.